Derivation of the Transformed Equation

- Let $t = \ln x$, so $x = e^t$ and $\dfrac{dt}{dx} = \dfrac{1}{x}$.
- By the chain rule: $\dfrac{dy}{dx} = \dfrac{dy}{dt}\dfrac{dt}{dx} = \dfrac{1}{x}\dfrac{dy}{dt}$
- Hence $x\dfrac{dy}{dx} = \dfrac{dy}{dt}$, often written $xy' = \dot{y}$
- For the second derivative, (use chain rule),

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dt}\right) = \frac{1}{x^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$$

- Hence $x^2\dfrac{d^2y}{dx^2} = \dfrac{d^2y}{dt^2} - \dfrac{dy}{dt}$, i.e. $x^2y'' = \ddot{y} - \dot{y}$
- Substitute these into $ax^2y'' + bxy' + cy = f(x)$:

$$a(\ddot{y} - \dot{y}) + b\dot{y} + cy \implies f(e^t) = a\ddot{y} + (b-a)\dot{y} + cy = f(e^t).$$

- Note the AE of this new ODE is $am(m-1) + bm + c = 0$ and it matches with the AE to the Cauchy–Euler derived earlier.
- So both methods must always agree, as expected



Correspondence Between the Two Methods

- Direct method: assume $y = x^m$, solve $am(m-1) + bm + c = 0$ for m.
- Transform method: substitute $t = \ln x$, solve the same auxiliary equation for the roots of t-equation.
- Distinct real roots m_1, m_2: direct gives x^{m_1}, x^{m_2}; transform gives $e^{m_1 t}, e^{m_2 t} = x^{m_1}, x^{m_2}$.
- Repeated root m: direct gives $x^m, x^m \ln x$; transform gives $e^{mt}, te^{mt} = x^m, x^m \ln x$.
- Again the two methods agree exactly, confirming the log-term rule.
- Complex roots $\alpha \pm i\beta$: transform gives $e^{\alpha t}\cos\beta t, e^{\alpha t}\sin\beta t$. which becomes $x^\alpha \cos(\beta \ln x), x^\alpha \sin(\beta \ln x)$ after back-substitution.
- Once again this matches the direct method exactly.
- The two methods are therefore mathematically equivalent.
- The transform method is usually preferred for non-homogeneous problems.



The Non-homogeneous Cauchy–Euler Equation

- The non-homogeneous form is

$$ax^2y'' + bxy' + cy = f(x), \qquad x > 0.$$

- As with any linear ODE, the general solution has two parts

$$y = y_c(x) + y_p(x).$$

- y_c is the complementary function (Solution of the homogeneous diff. equation).
- y_c is found exactly as in Cases 1, 2, 3 discussed earlier.
- y_p is any one particular solution of the full non-homogeneous equation.
- Finding y_p is the new challenge specific to non-homogeneous problems
- The choice of technique depends on the form of $f(x)$.
- So y_c must always be computed before attempting y_p.



Two Techniques for Finding the Particular Solution

- **Technique A: Variation of Parameters**
- Works for any continuous $f(x)$, regardless of its form.
- Requires knowing both complementary solutions y_1, y_2 and their Wronskian.
- Involves computing two integrals, which can sometimes be difficult.
- **Technique B: Transform to $t = \ln x$, then use Undetermined Coefficients**
- Works only when $f(x)$ becomes a "nice" function of t after substitution.
- For example, $f(x) = x^k$ becomes $f(e^t) = e^{kt}$, a standard UC-friendly form.
- Similarly $f(x) = x^k \ln x$ becomes te^{kt}, still UC-friendly.
- After finding $y_p(t)$, substitute back $t = \ln x$ to express y_p in terms of x.
- Technique B is usually faster when $f(x)$ is a power of x (with or without $\ln x$).



Variation of Parameters Formula

- Write the equation in standard form by dividing by ax^2:

$$y'' + \frac{b}{ax}y' + \frac{c}{ax^2}y = \frac{f(x)}{ax^2} =: g(x)$$

- Let y_1, y_2 be the two independent complementary solutions.
- Let $W(x) = y_1y_2' - y_2y_1'$ be their Wronskian.
- The particular solution is given by

$$y_p = -y_1 \int \frac{y_2\, g(x)}{W(x)}\, dx \;+\; y_2 \int \frac{y_1\, g(x)}{W(x)}\, dx$$

- This formula is completely general and always applies to linear second order ODEs.
- The main work is computing the two indefinite integrals.
- For Cauchy–Euler equations, $W(x)$ is always a simple power of x.
- This often makes the integrals more tractable than in general variable-coefficient problems.
- Once y_p is found, add it to y_c for the full general solution.



Example 1: Distinct Real Roots (Setup)

- Solve the homogeneous equation: $x^2y'' - 2xy' - 4y = 0$, $x > 0$
- Here $a = 1, b = -2, c = -4$.
- Try $y = x^m$, so $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$.
- Substitute into the ODE:

$$x^2 \cdot m(m-1)x^{m-2} - 2x \cdot mx^{m-1} - 4x^m = 0.$$

- Simplify: $\big[m(m-1) - 2m - 4\big]x^m = 0.$
- Auxiliary equation: $m(m-1) - 2m - 4 = 0.$
- Expand: $m^2 - m - 2m - 4 = 0 \implies m^2 - 3m - 4 = 0$
- Factor: $(m-4)(m+1) = 0.$
- Roots: $m_1 = 4,\ m_2 = -1.$
- These are distinct real roots, so Case 1 theory applies directly.



Example 1: Solution

- Roots found: $m_1 = 4$ and $m_2 = -1$
- By Case 1 theory, the two independent solutions are $y_1 = x^4$ and $y_2 = x^{-1}$
- General solution:
$$y = C_1 x^4 + C_2 x^{-1}$$
- Check: substitute $y_1 = x^4$ back into the ODE to verify it is indeed a solution
- $y_1' = 4x^3,\ y_1'' = 12x^2$; then
$x^2(12x^2) - 2x(4x^3) - 4x^4 = 12x^4 - 8x^4 - 4x^4 = 0$ ✓
- Check: for $y_2 = x^{-1}$: $y_2' = -x^{-2}, y_2'' = 2x^{-3}$;
$x^2(2x^{-3}) - 2x(-x^{-2}) - 4x^{-1} = 2x^{-1} + 2x^{-1} - 4x^{-1} = 0$ ✓
- Both checks confirm the solution is correct.
- As $x \to 0^+$, the term $C_2 x^{-1}$ dominates and blows up unless $C_2 = 0$.
- As $x \to \infty$, the term $C_1 x^4$ dominates.



Example 2: Repeated Root (Setup)

- Solve: $x^2y'' - 3xy' + 4y = 0, \quad x > 0.$
- Here $a = 1, b = -3, c = 4.$
- Try $y = x^m$, so $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$.
- Substitute into the ODE:

$$x^2 \cdot m(m-1)x^{m-2} - 3x \cdot mx^{m-1} + 4x^m = 0.$$

- Simplify: $\big[m(m-1) - 3m + 4\big]x^m = 0.$
- Auxiliary equation: $m(m-1) - 3m + 4 = 0.$
- Expand: $m^2 - m - 3m + 4 = 0 \implies m^2 - 4m + 4 = 0.$
- Factor: $(m-2)^2 = 0.$
- Repeated root: $m_1 = m_2 = 2.$
- Since the discriminant is exactly zero, this is Case 2 theory.



Example 2: Solution

- Repeated root found: $m_1 = m_2 = 2$.
- By Case 2 theory, the two independent solutions are $y_1 = x^2$ and $y_2 = x^2 \ln x$.
- General solution:

$$y = C_1 x^2 + C_2 x^2 \ln x = x^2(C_1 + C_2 \ln x).$$

- Verification: $y_2' = 2x \ln x + x, \quad y_2'' = 2 \ln x + 3$
- Substitute into the ODE:
  $x^2(2 \ln x + 3) - 3x(2x \ln x + x) + 4x^2 \ln x$
- $= 2x^2 \ln x + 3x^2 - 6x^2 \ln x - 3x^2 + 4x^2 \ln x = 0$ ✓
- This confirms y_2 genuinely solves the equation
- As $x \to 0^+$, $x^2 \ln x \to 0$, so both solutions vanish at the origin.
- As $x \to \infty$, both solutions grow, with y_2 growing slightly faster.



Example 3: Complex Roots (Setup)

- Solve: $x^2y'' + xy' + y = 0, \quad x > 0$
- Here $a = 1, b = 1, c = 1$.
- Try $y = x^m$, so $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$.
- Substitute into the ODE and simplify to get

$$\big[m(m-1) + m + 1\big]x^m = 0.$$

- Auxiliary equation: $m(m-1) + m + 1 = 0$.
- Simplify: $m^2 - m + m + 1 = 0 \implies m^2 + 1 = 0$.
- Roots: $m = \pm i$.
- Write $\alpha = 0$ (real part) and $\beta = 1$ (imaginary part).
- Since roots are complex conjugates, Case 3 theory applies.



Example 3: Solution

- Roots found: $m = \pm i$, so $\alpha = 0,\ \beta = 1$
- By Case 3 theory, the two real independent solutions are

$$y_1 = \cos(\ln x), \qquad y_2 = \sin(\ln x).$$

- General solution:

$$y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$$

- Verification (sketch): differentiate y_1 twice using the chain rule and substitute back
- $y_1' = -\dfrac{\sin(\ln x)}{x}, \quad y_1'' = \dfrac{-\cos(\ln x) + \sin(\ln x)}{x^2}.$
- Substituting confirms the ODE $x^2y'' + xy' + y = 0$ is satisfied.
- As $x \to 0^+$, $\ln x \to -\infty$, so the solution oscillates infinitely often.
- As $x \to \infty$, the solution also oscillates without any growth or decay.
- Neither solution grows or decays in amplitude since $\alpha = 0$. .



Example 4: Non-homogeneous, Power of x (Setup)

- Solve: $x^2y'' - 2xy' - 4y = x^3, \quad x > 0$
- The complementary equation is exactly Example 1: $y_c = C_1x^4 + C_2x^{-1}$
- We now find a particular solution y_p using the transform method.
- Let $t = \ln x$, so $x = e^t$; recall $xy' = \dot{y}$ and $x^2y'' = \ddot{y} - \dot{y}$
- The transformed equation becomes

$$\ddot{y} - \dot{y} - 2\dot{y} - 4y = e^{3t} \implies \ddot{y} - 3\dot{y} - 4y = e^{3t}$$

- The homogeneous part has auxiliary equation $m^2 - 3m - 4 = 0$, roots $4, -1$ (matches Example 1)
- Since RHS is e^{3t} and 3 is not a root, try $y_p = Ae^{3t}$.
- Substitute: $9A - 9A - 4A = A(9 - 9 - 4) = -4A$.
- Set $-4A = 1 \Rightarrow A = -\frac{1}{4}$.
- So $y_p(t) = -\frac{1}{4}e^{3t}$.
- Convert back to x: since $e^{3t} = x^3$, we get $y_p(x) = -\frac{1}{4}x^3$.



Example 4: Final Solution

- From the previous slide, the particular solution is $y_p(x) = -\frac{1}{4}x^3$.
- Quick check: substitute $y_p = -\frac{1}{4}x^3$ directly into the original ODE.
- $y_p' = -\frac{3}{4}x^2$, $y_p'' = -\frac{3}{2}x$.
- $x^2(-\frac{3}{2}x) - 2x(-\frac{3}{4}x^2) - 4(-\frac{1}{4}x^3) = -\frac{3}{2}x^3 + \frac{3}{2}x^3 + x^3 = x^3$ ✓
- This confirms y_p satisfies the non-homogeneous equation exactly.
- The general solution is given by

$$y = C_1x^4 + C_2x^{-1} - \frac{1}{4}x^3$$

- This illustrates Technique B: transform, use undetermined coefficients, transform back.
- The method was much faster here than variation of parameters would have been.
- This works well whenever $f(x)$ is a pure power of x.



Example 5: Non-homogeneous via Variation of Parameters (Setup)

- Solve: $x^2y'' - 2xy' + 2y = x^3 \ln x, \quad x > 0$
- First find y_c: try $y = x^m$, auxiliary equation $m(m-1) - 2m + 2 = 0$.
- Simplify: $m^2 - m - 2m + 2 = 0 \Rightarrow m^2 - 3m + 2 = 0 \Rightarrow (m-1)(m-2) = 0$.
- Roots: $m_1 = 2, \ m_2 = 1$ (distinct real roots)
- Complementary function: $y_c = C_1x^2 + C_2x$
- Convert the ODE to standard form by dividing by x^2:

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = x \ln x$$

- So $g(x) = x \ln x$ is the right-hand side in standard form.
- Take $y_1 = x^2$, $y_2 = x$.
- Compute the Wronskian: $W = y_1y_2' - y_2y_1' = x^2(1) - x(2x) = x^2 - 2x^2 = -x^2$
- We are now ready to apply the variation of parameters



Example 5: Computing y_p and y

- Variation of parameters formula:

$$y_p = -y_1 \int \frac{y_2\, g(x)}{W}\, dx + y_2 \int \frac{y_1\, g(x)}{W}\, dx$$

- First integral: $\displaystyle\int \frac{x \cdot x \ln x}{-x^2}\, dx = -\int \ln x\, dx = -(x \ln x - x)$
- Second integral:
$\displaystyle\int \frac{x^2 \cdot x \ln x}{-x^2}\, dx = -\int x \ln x\, dx = -\left(\frac{x^2}{2} \ln x - \frac{x^2}{4}\right)$
- $y_p = -x^2\big[-(x \ln x - x)\big] + x\left[-\left(\frac{x^2}{2} \ln x - \frac{x^2}{4}\right)\right]$
- Simplifying : $y_p = \dfrac{x^3}{2} \ln x - \dfrac{3x^3}{4}$.
- General solution:

$$y = C_1 x^2 + C_2 x + \frac{x^3}{2} \ln x - \frac{3x^3}{4}$$



Example 6: Full Transform-Method Example (Setup)

- Solve: $x^2y'' + xy' - y = x^2, \quad x > 0$
- First find y_c: auxiliary equation
  $m(m-1) + m - 1 = 0 \Rightarrow m^2 - 1 = 0.$
- Roots: $m_1 = 1,\ m_2 = -1$ (distinct real roots).
- Complementary function: $y_c = C_1x + C_2x^{-1}$.
- Now transform using $t = \ln x$: recall $xy' = \dot{y}$, $x^2y'' = \ddot{y} - \dot{y}$
- Transformed equation:
  $(\ddot{y} - \dot{y}) + \dot{y} - y = e^{2t} \implies \ddot{y} - y = e^{2t}$
- Homogeneous part has roots ± 1 (matches y_c above, as expected)
- Since RHS is e^{2t} and 2 is not a root, try $y_p = Ae^{2t}$.
- Substitute: $4A - A = 3A$; set $3A = 1 \Rightarrow A = \frac{1}{3}$
- So $y_p(t) = \frac{1}{3}e^{2t}$.



Example 6: Final Solution

- Since $e^{2t} = x^2$, we get $y_p(x) = \frac{1}{3}x^2$.
- Verify directly: $y_p' = \frac{2}{3}x$, $y_p'' = \frac{2}{3}$.
- Substitute: $x^2(\frac{2}{3}) + x(\frac{2}{3}x) - \frac{1}{3}x^2 = \frac{2}{3}x^2 + \frac{2}{3}x^2 - \frac{1}{3}x^2 = x^2$ ✓
- This confirms the particular solution is correct.
- General solution:

$$y = C_1 x + C_2 x^{-1} + \frac{1}{3}x^2$$

- This example again shows Technique B is fast whenever $f(x)$ is a pure power of x.
- Comparing Examples 4, 5, and 6 shows both techniques in complementary roles.
- Technique B is preferred for powers of x (with or without $\ln x$).

